Real time Accommodation Measurement

Abstract

CCS Concepts

• **Computing methodologies → Image compression**; • **Theory of computation → Data compression**; • **Hardware →** *Displays and imagers*; *Emerging optical and photonic technologies.*

Keywords

eyes tracking, deep learning, computer graphics

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1 Introduction

2 Related works

2.1 Tracking eye and gaze

In stereoscopic displays, the Vergance-Accommodation Conflict (VAC) arises because the eyes converge to the depicted 3D depth while accommodation and retinal blur remain tied to the physical display surface, which can impair fusion and cause visual discomfort. [Hoffman et al. 2008] Recent VAC-related depth-perception studies therefore use eye-tracking and controlled optical setups to measure vergence responses under different perceived depths, highlighting the need for objective and reproducible eye-based measurements in VAC-related experiments [Bontha and Arefin 2024; Sturgeon et al. 2025]. Therefore, Purkinje-image-based methods seek to recover accommodative cues from intraocular reflections: Lu et al.[2020] used a multi-camera AR-glasses prototype to track higher-order Purkinje images and reported a best vergence error of 0.3782° and an accommodation-depth error of 1.108 cm [Lu et al. 2020]. Ozhan et al.[2023]further use a Purkinje imaging setup with a MLP-based regression model, achieving 0.17 D accommodation RMSE and 0.18° vergence RMSE with subject-specific data, and 0.32 D / 0.32° with two-point calibration [Ozhan et al. 2023]. These works suggest that higher-order Purkinje reflections can serve as physically meaningful cues for accommodation-aware eye tracking, but their weak intensity, defocus, and strong dependence on LED–camera–eye geometry make them difficult to capture robustly. Motivated by this limitation, we first build a controllable Blender/LuxCore optical eye model to analyze the formation, visibility, and stability of P1–P4 under near-eye infrared imaging conditions.

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Table 1: Comparison of representative eye-tracking, Purkinje-based accommodation measurement, and near-eye modeling systems.

Work	Camera / illumination	Mode	Reported accuracy / result
Cornsweet and Crane [1973]	Dual-Purkinje optical tracker; infrared illumination	First/fourth Purkinje eye-position tracking	∼ 1 arcmin eye-position accuracy
Morimoto et al. [2002]	Single camera; at least two NIR light sources	Remote gaze tracking with Gullstrand eye model	∼ 5° gaze error at 500–600 mm
Sturgeon et al. [2025]	Integrated eye tracker; OST AR display	Vergence response measurement under perceived-depth changes	EVA response analysis
Bontha and Arefin [2024]	Controlled AR optical setup	Replicated AR haploscope for depth presentation	Reproducible optical setup
Ozhan et al. [2023]	Single NIR camera; one 940 nm NIR LED	Purkinje-based vergence and accommodation measurement	0.18° vergence / 0.17 D accommodation
Lu et al. [2020]	Multiple cameras; multiple IR LEDs	Near-eye AR Purkinje tracking	0.3782° vergence; 1.108 cm accommodation-depth error
RIT-Eyes [2020]	Synthetic camera; Blender/Cycles illumination	Synthetic near-eye image generation	86.06 global mIoU for eye-region segmentation
EyeNeRF [2022]	Sparse cameras and lights; handheld camera with co-located light	Hybrid mesh–NeRF eye reconstruction and relighting	Photorealistic eye synthesis / relighting
Khademi et al. [2026]	Sparse multi-view cameras; point lights/fringe projector	Relightable mesh–NeRF eye reconstruction	Realistic relightable eye reconstruction

2.2 Reconstructive surgery of the eyeball with adjustable lighting and end-to-end physical simulation

Excerp for physical capture, for the optical eyes modeling, the main problem including the complicated geometry of outsurface of cornea. For example, eyenerf propose a hybrid representation, conbine explicit eyeball model to handle high-frequency high-gloss , in the mean time, Training the Nerf model for implicit representation enables the modeling of the surrounding eye area and the complex appearance within the eye[Li et al. 2022]. In addition,

Reconstructing Realistic and Relightable Eyes highlight the importance of near-field illuminationb to eyes modelling this use hybrid mesh + NeRF Rebuild the ocular system with reflective light capabilities and support the common near-field point lights and structured/fringe projectors [Khademi et al. 2026].

Differentiable optical design provides a useful reference for future optimization of the LED–camera layout in our system. *End-to-end Optimization of Fluidic Lenses* integrates fluidic lens geometry, differentiable ray tracing, and neural reconstruction into a single end-to-end pipeline, allowing the loss to be back-propagated to lens design parameters [Na et al. 2024]. Similarly, *End-to-End Hybrid Refractive-Diffractive Lens Design with Differentiable Ray-Wave Model* combines ray tracing with wave propagation to jointly optimize refractive-diffractive optics and the reconstruction network [Yang et al. 2024]. These works suggest that optical simulation can be part of the design loop; in our case, future optimization could use the brightness, sharpness, separation from P1, and field-of-view coverage of P3/P4 as objectives for LED and camera placement.

In real captures, higher-order Purkinje images, especially P3 and P4, are relatively faint and sensitive to defocus and sensor noise. Related enhancement and adaptation methods suggest possible post-processing tools: unsupervised anti-noise computational imaging reconstructed 64×64 images at 20 fps with only a 5% sampling ratio [Zhai et al. 2022], diffusion-based style transfer reduced synthetic-to-real FID to 54.56 compared with 65.31 and 73.26 for prior baselines [Chigot et al. 2025], and physics-informed diffusion reduced physical residuals by about two orders of magnitude in a Darcy-flow task and by 33%–35% in topology optimization [Bastek et al. 2025].

3 Background

3.1 Optical basis of Purkinje images and VAC

In natural vision, vergence and accommodation are coupled: the eyes rotate toward a target while the crystalline lens changes optical power to focus it on the retina. In AR/VR displays, virtual depth cues may drive vergence to a distance different from the display focal plane, causing the vergence–accommodation conflict (VAC). As holographic and metasurface-waveguide AR displays move toward more realistic 3D imagery, measuring both vergence and accommodation becomes important for evaluating depth perception and visual comfort [Gopakumar et al. 2024].

Conventional eye trackers estimate gaze or vergence from pupil, iris, limbus, or corneal-glint features, which mainly describe eye rotation and provide limited access to accommodation [Nair et al. 2020]. Purkinje images offer a more direct optical cue: P1/P2 are mainly associated with the cornea, while P3/P4 originate from the crystalline lens surfaces and are related to accommodation [Lu et al. 2020][Ozhan et al. 2023]. Because P3/P4 are faint, easily defocused, and sensitive to LED–camera–eye geometry, we first study their formation in a controllable optical eye model before real IR capture.

3.2 Simulation-to-capture pipeline for near-eye Purkinje imaging

A reliable Purkinje imaging pipeline requires the eye to be modeled as an optical system rather than an appearance object. Since P1–P4 are generated by reflection and refraction at corneal and lenticular surfaces, Arizona/ZEMAX-style schematic eye models are useful because they make accommodation, refractive interfaces, and ray paths physically controllable [Lu et al. 2020][Ozhan et al. 2023]. This also explains why eye-rendering methods often keep the eyeball or cornea as an explicit mesh: specular highlights, corneal refraction, and near-field glints are tied to surface geometry, while NeRF-like fields are more suitable for view synthesis, relighting, and periocular appearance modeling [**???**].

The same optical constraints determine capture quality. LED and camera placement must preserve a stable eye box while avoiding eyelid/eyelash occlusion, glint overlap, saturation, and defocus; otherwise weak P3/P4 reflections can merge with P1, fall outside the field of view, or become unresolvable at the sensor resolution. Thus, the key background principle is that eye geometry, near-field illumination, camera focus/FOV, resolution, and reconstruction representation must be designed together before reliable Purkinje capture and later NeRF/3D Gaussian reconstruction can be expected [Kerbl et al. 2023]

4 Results and Discussions

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